

PlantForge.ai — demo script

Problem Statement 8 · AI for Industrial Knowledge Intelligence: Unified Asset

& Operations Brain

Runtime: ~7:15 · Narration: ~1,050 words at 150 wpm

Record the screen **silent**, generate the voice-over from the NARRATION blocks, then cut the video under the audio. Fitting picture to a fixed audio track is far easier than syncing narration to clips already timed.

Pre-flight

- **Log in as an admin.** Admin is the only role that reveals the whole navigation, including Connectors — the walkthrough depends on it.
- **Regenerate the alerts.** Those currently on the feed were written under an older prompt: they open with a stray rule and claim a work order was "drafted in SAP", which is untrue. New investigations produce three clean sections ending in a numbered checklist.
- **Pre-warm every page once** — first load of Graph Explorer is ~850 ms and the answer cache starts cold. Clicking through beforehand makes the recording feel instant.
- **Have the Ask question on the clipboard**, not typed live.
- Browser at 1920×1080, 100% zoom, bookmarks hidden, single tab.

For the voice tool

Feed it the NARRATION text only — no markdown. Equipment tags read badly as literals, so spell them phonetically in the input: P-101A → "P one oh one A", K-301 → "K three oh one", OISD → "O I S D", P&ID → "P and I D".

Scene 1 — The problem, and the statement (0:00–0:50)

SCREEN: Landing page. Slow scroll across the statistics band, hold on it.

NARRATION

A 2024 McKinsey survey found that professionals in asset-intensive industries spend thirty-five percent of their working hours searching for information that already exists somewhere in their own organisation. In Indian heavy industry, a large plant runs on seven to twelve disconnected document systems — drawings in one, work orders in another, procedures in a third, inspection records in a fourth, and regulatory correspondence scattered across email. That fragmentation drives an estimated eighteen to twenty-two percent of unplanned downtime. And within the next decade, a quarter of India's experienced engineers retire and take decades of undocumented knowledge with them. That is Problem Statement Eight. We chose it because it is not a document management problem. It is a safety problem, a compliance problem and a reliability problem wearing a filing cabinet as a disguise. This is PlantForge.

Scene 2 — The unified brain (0:50–1:40)

SCREEN: Sign in as admin → Dashboard. Let the KPI row and charts settle. Hover the admin role badge. Then sweep the full left navigation slowly, top to bottom.

NARRATION

We are signed in as an administrator, so every capability is visible. The platform ingests engineering drawings, maintenance records, procedures, inspection reports and email archives, and turns them into one connected knowledge graph — currently a little over two thousand entities linked across forty-two documents. This dashboard is not a mock-up. Every number here is read live from that graph and from the running pipeline: alerts the agents raised, where the ingestion queue stands, and what the plant is exposed to right now. What you see in the sidebar is one brain, addressed by different jobs — asking it a question, watching what it warns you about, planning a change, proving compliance, and capturing what is still only in someone's head.

Scene 3 — Ask anything, and check the answer (1:40–2:45)

SCREEN: Ask → paste *"Why does P-101A keep losing its mechanical seal?"* Let it stream. **Click a citation chip** — hold on the opened source document for three full seconds. Close.

NARRATION

The first job is the one everybody wants: ask the plant a question in plain language and get a real answer. This is retrieval over the whole corpus at once — work orders, procedures and incident reports that no single search box spans today. But the answer is not the interesting part. This is. Every claim carries a citation, and a citation is not a footnote — it opens the source. That is the actual standard operating procedure the answer relied on, retrieved from storage, exactly as it was filed. The system also reports its own confidence, based on how completely the answer traced back to evidence. If it cannot ground a statement in a document you own, it says so instead of sounding certain. No engineer signs off on an answer they cannot check, so we made every answer checkable.

Scene 4 — The graph underneath (2:45–3:20)

SCREEN: Graph Explorer. One slow pan across the network. Hover a node or two to surface labels. Do not click repeatedly.

NARRATION

Underneath is why this answers questions a search engine cannot. Documents were not just indexed — they were read, and the things inside them extracted as entities: equipment tags, failure modes, procedures, work orders, regulatory clauses and the people involved. Then they were linked. This pump is connected to that vessel, governed by that standard, repaired under that procedure, mentioned in those four documents. When a question needs a connection that spans three documents and two decades, it is already here, waiting.

Scene 5 — Failure intelligence, unprompted (3:20–4:20)

SCREEN: Alerts. Open the **P-101A seal-leak** alert. Scroll slowly so the three sections and the numbered first-checks list are all shown. Hover the source chips.

NARRATION

Nobody asked for this one. When new failure data lands, an agent investigates on its own — the asset's own history, its sibling equipment, the procedures that fixed it before, and everything it is physically connected to. Here it concluded that P one oh one A is not failing because of a defective seal. It is failing from suction starvation — the same signature its sibling pump shows — and a design review recommended a year ago was never actioned. It then lists the specific checks to make before that pump goes back into service. That conclusion required connecting four separate documents filed by four different people in four different systems. No individual had read all four. This is the lessons-learned problem the statement describes, solved continuously instead of during the post-incident review.

Scene 6 — From a warning to scheduled work (4:20–5:05)

SCREEN: Click **A corrective work order was drafted from this** → Work Orders. Point out the priority badge, then the harvested fact chips, then the grounded banner. Click **Approve**, show the approved state.

NARRATION

A warning that stops at a warning is just noise, so the same investigation drafts the corrective work order. Look at how it is assembled. The affected equipment, the prior work orders and the procedures are lifted directly out of the graph — the model is not permitted to write into those lists. Only the two narrative paragraphs come from the model, and they are checked against the evidence before anyone sees them. The priority is calculated from recurrence and regulatory exposure, not chosen by an AI. And nothing here is automatic. It stays a draft until a planner approves it, because committing a technician's shift is a person's decision, not a model's.

Scene 7 — Compliance you can prove (5:05–5:50)

SCREEN: Compliance. Let the counts land. Expand an **overdue** item → **View evidence** (document opens) → close → **Schedule inspection**. Then flick to Work Orders to show the new preventive draft.

NARRATION

Regulatory obligations come out of the same graph. Seven statutory inspections currently overdue, against real standards — O I S D and pressure-vessel codes — mapped onto the specific assets they govern. View evidence opens the document the obligation was read from, which is exactly what an auditor asks for. Scheduling one does not quietly book anything. It drafts a preventive work order into the same approval queue as everything else, because compliance does not get a side door into the maintenance schedule.

Scene 8 — Before you change anything (5:50–6:25)

SCREEN: Change Impact. Use the **V-203** example. Let the evidence steps tick through live, then the assessment.

NARRATION

Before a modification is made, you can ask what it touches. Watch the left side — that is the agent walking the plant's real topology, one query at a time: connections, failure history, governing clauses, and every document that would need revising. What comes back is not a verdict on whether to proceed. It is the evidence a competent person needs in order to decide, assembled in seconds instead of over a fortnight.

Scene 9 — Catching the knowledge before it walks out (6:25–7:15)

SCREEN: Interview. Start a session, ask one or two questions, show the live transcript. Then show the generated handover document with its structure.

NARRATION

Which leaves the hardest part of the statement — the quarter of India's experienced engineers retiring within the decade. You cannot solve that by asking someone to write documentation. So the system interviews them. It reads the graph first, so it already knows which assets this person worked on and what has historically failed on them, and it asks about those specifically — the judgement that never reached a document. Why this pump gets a different seal. What the vibration actually means before the alarm. What comes out is a structured handover document, and it goes straight back through the same ingestion pipeline as any drawing or work order. So the moment the conversation ends, that knowledge is answerable in Ask, citable in an assessment, and available to whoever replaces them. Thirty-five percent of the working week, seven to twelve disconnected systems, and a retirement cliff — one graph, one brain, every answer traceable to the document it came from. That is PlantForge.

If you need a shorter cut

Drop **Scene 8 (Change Impact)** first, then **Scene 4 (Graph)** — that lands around 5:30 and keeps the problem statement, the citations, the failure intelligence, compliance and the retirement cliff.

Do **not** drop Scene 3's citation click or Scene 9. Those are the two moments that separate this from a chatbot over a document folder.

Lines worth protecting

- *"A safety problem, a compliance problem and a reliability problem wearing a filing cabinet as a disguise."*
- *"A citation is not a footnote — it opens the source."*
- *"No individual had read all four."*
- *"The model is not permitted to write into those lists."*
- *"You cannot solve that by asking someone to write documentation. So the system interviews them."*

